

Jack Hollis Farrell

Department of Physics
Center for Theory of Quantum Matter
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EDUCATION

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| <i>Ph.D. Physics</i> , University of Colorado Boulder | 2021–2027 |
| • Advisor: Prof. Andrew Lucas | |
| <i>Honours B.Sc. Physics</i> , University of Toronto | 2017–2021 |
| • Major in Physics, Minors in Mathematics and English. | |

PUBLICATIONS

Publications and Preprints

1. **Jack H. Farrell** and Andrew Lucas. “Characterizing electronic scattering rates with transport in multiterminal devices”, [arXiv: 2605.03030 \(2026\)](#).
2. Ludwig Holleis, Youngjoon Choi, Canxun Zhang, **Jack H. Farrell**, Gabriel Bargas, Audrey Hsu, Zexing Chen, Ian Sackin, Wenjie Zhou, Yi Guo, Thibault Charpentier, Yifan Jiang, Benjamin A. Foutty, Aidan Keough, Martin E. Huber, Takashi Taniguchi, Kenji Watanabe, Andrew Lucas, and Andrea F. Young. “Cryogenic shock exfoliation for ultrahigh mobility rhombohedral graphite nanoelectronics”, [arXiv: 2604.21912 \(2026\)](#).
3. Canxun Zhang, Evgeny Redekop, Hari Stoyanov, **Jack H. Farrell**, Sunghoon Kim, Ludwig Holleis, David Gong, Yongjoon Choi, Takashi Taniguchi, Kenji Watanabe, Martin E. Huber, Ania C. Bleszynski Jayich, Andrew Lucas, and Andrea F. Young. “Imaging electron hydrodynamics in a graphene flat band system”, [arXiv: 2603.11175 \(2026\)](#).
4. Johannes Geurs, Tatiana A. Webb, Yinjie Guo, Itai Keren, **Jack H. Farrell**, Jikai Xu, Kenji Watanabe, Takashi Taniguchi, Dmitri N. Basov, James Hone, Andrew Lucas, Abhay Pasupathy, Cory R. Dean. “Supersonic flow and hydraulic jump in an electronic de Laval nozzle”, [arXiv: 2509.16321 \(2025\)](#).
5. Xiaoyang Huang, **Jack H. Farrell**, Aaron J. Friedman, Isabella Zane, Paolo Glorioso, and Andrew Lucas. “Generalized time-reversal symmetry and effective theories for nonequilibrium matter”, [arXiv: 2310.12233 \(2023\)](#).
6. **Jack H. Farrell**, Xiaoyang Huang, and Andrew Lucas. “Hydrodynamics with helical symmetry”, [arXiv: 2208.12269 \(2022\)](#).
7. **Jack H. Farrell**, Nicolas Grisouard, and Thomas Scaffidi. “Terahertz radiation from the Dyakonov-Shur instability of hydrodynamic electrons in a Corbino geometry”. *Phys. Rev. B* **106**, 195432 (2022).

Manuscripts in Preparation

1. **Jack H. Farrell**, Akshay Pal, and Andrew Lucas. “Hydrodynamic-to-ballistic crossover of electron transport in complex devices”. Manuscript in preparation (2026).
2. Akshay Pal, Andrew Lucas, and **Jack H. Farrell**. “Electron hydrodynamics with an open Fermi surface”. Manuscript in preparation (2026).
3. Elijah G. Lew-Smith, Marvin Qi, **Jack H. Farrell**, and Andrew Lucas. “A lattice model of constant density flocking without long range order”. Manuscript in preparation (2026).
4. **Jack H. Farrell** and Andrew Lucas. “Endpoint of a ballistic Dyakonov-Shur instability”. Manuscript in preparation (2026).

PRESENTATIONS

Invited Talks

1. “[Computational Fluid Mechanics for Electrons](#)”. Centre de Physique Théorique, École Polytechnique (2026).
2. “[Compressible Flow Instabilities in Electronic Liquids](#)”. Cory Dean Group Meeting, Columbia University (2025).
3. “[Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter](#)”. CTQM Seminar, Center for Theory of Quantum Matter (2024).

Contributed Talks

1. “[2D Electron Transport in Complex Geometries](#)”. Correlated Quantum Systems Workshop, Institut Pascal (2026). [Video recording](#).
2. “[2D Electron Transport through Complex Geometries](#)”. APS Global Physics Summit (2026).
3. “[Endpoint of a Ballistic Dyakonov-Shur Instability](#)”. APS March Meeting (2025).
4. “[Constant Density Flocking without Long-Range Order](#)”. Quantum, Classical, and Active Fluids Workshop, CUNY Graduate Center (2024). [Video recording](#).
5. “[Effective Theories for Nonequilibrium Matter](#)”. APS March Meeting (2024).
6. “[Hydrodynamics with Helical Symmetry](#)”. APS March Meeting (2023).

Poster Presentations

1. “[Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter](#)”. UQM: Ultra Quantum Matter (2024).

HONOURS

NSERC Canada Graduate Scholarship (CGS-M), NSERC 2021

- Declined: offered at U of T, UBC, University of Waterloo / Perimeter Institute.
- Research grant for first-year Ph.D. students.

University of Toronto National Scholarship 2017–2021

- Full scholarship to U of T, covering tuition and living fees for four years.

<i>NSERC Undergraduate Summer Research Award</i> , University of Toronto	2020
<i>Hymie and Roslyn Mida Award in Theoretical Physics</i> , University of Toronto	2020
<i>NSERC Undergraduate Summer Research Award</i> , University of Toronto	2019
<i>University of Toronto Scholar</i>	2018–2019

TEACHING

<i>Instructor of Record</i> , University of Colorado Boulder	
• PHYS 1400: Fundamentals of Scientific Inquiry (course webpage)	2025
<i>Guest Lecturer</i> , University of Colorado Boulder	
• PHYS 7810: Graduate Special Topics: Hydrodynamics as an Effective Field Theory	2026
• PHYS 5210: Graduate Theoretical Mechanics	2024
• PHYS 4410: Quantum Mechanics II	2023
<i>Teaching Assistant</i> , University of Colorado Boulder	
• PHYS 7810: Graduate Special Topics in Physics: Hydrodynamics	Spring 2024
• PHYS 3090: Introduction to Quantum Computing	Spring 2024
• PHYS 1110: General Physics I	Fall 2023
• PHYS 1125: General Physics II for Majors	Spring 2023
• PHYS 1110: General Physics I	Fall 2022
• PHYS 1110: General Physics I	Spring 2022
• PHYS 1110: General Physics I	Fall 2020
<i>Teaching Assistant</i> , University of Toronto	2020–2021
• Extra help for programming in <code>Python</code> for physics majors	

SERVICE

<i>Program Assistant</i> , Boulder Summer School for Condensed Matter and Materials Physics	
• Geometry and Topology in Soft Matter	2026
• Dynamics of Strongly Correlated Electrons	2025
• Self-Organizing Matter: From Inanimate to the Animate	2024
• Non-Equilibrium Quantum Dynamics	2023
• Hydrodynamics Across Scales	2022

MENTORING & SUPERVISION

<i>Research mentoring</i> , University of Colorado Boulder	
• Akshay Pal, CU Boulder, graduate student	2025–present
• Julia Gao, Boulder High School	2026–2027
• Zachary Hill, UT Austin REU	2026–2027
• Andrew Hicks, CU Boulder, undergraduate (now Ph.D. student, Yale University)	2022–2024
• Elijah G. Lew-Smith, Brown U. REU (now Ph.D. student, MIT)	2023–2024